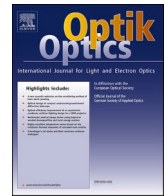




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Original research article

Remote sensing image scene classification using CNN-MLP with data augmentation

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ABSTRACT

Classification of the very high-resolution (VHR) imagery scene has become a challenging problem. The convolutional neural network (CNN) has increased the accuracy in this area due to learning features. However, models based on CNN contain many deep layers for classifying images that are not perfect in describing the relationship between objects within the image. Therefore, an enhanced multilayer perceptron (MLP) depending on Adagrad optimizer is employed in the classification step in this paper as a deep classifier. Motivated by this idea, this paper proposes an effective classification model named CNN-MLP to utilize the benefits of these two techniques: CNN and MLP. The features are generated using pre-trained CNN without fully connected layers. Due to limited training images per class, the proposed model uses data augmentation techniques to expand the training images. Then, an MLP is used to classify the final feature maps into the specified classes. Three public remote sensing datasets of VHR images to evaluate the proposed CNN-MLP model: UC-Merced, Aerial Image (AID), and NWPU-RESISC45 datasets. The experiment's findings show that the proposed method will contribute to higher classification performance relative to state-of-the-art methods.

1. Introduction

Nowadays, remote sensing of the Earth's surface is significantly contributing to explore the Earth and improving many applications related to learning about the features of these images. Many different types of remote sensing images are available now, e.g. multi/hyper-spectral, very high-resolution (VHR) images, and synthetic aperture radar (SAR). Thus, many techniques for the study of these types of images are presented in the last decade include remote sensing image identification, detection [1–5], compression [6–10], and classification [11–18]. In this research, we are concerned with VHR image classification. VHR image is one of the best surface observation data sources. Hence, it is important to understand its entities. VHR remote sensing image scene classification whose goal is to identify images based on its entities has a great benefit in the research area of remote sensing as there are essential applications in urban planning, managing land resources, tracking of disasters and traffic monitors [19].

Recently, the spread of deep learning methods has led to remarkable performance on remote sensing image classification. Owing to the great success achieved by convolutional neural networks (CNNs) in the computer vision community, a considerable number of

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CNN-based models have been proposed for remote sensing classification [20–24]. These models give classification performance better than other traditional methods. Moreover, in [25], an unsupervised learning method has been proposed to investigate de-convolution networks for remote sensing scene classification. This method works as follows: a shallow weighted convolution network learns a set of feature maps and filters for each image based on reducing the error between the input image and the output. On another research, CNN, as the feature extractor and linear support vector machine (SVM), is used as the final classifier in [26]. In addition, Zhu et al. [27] have introduced a model based on the fully sparse semantic by extracting heterogeneous low-level features, then performing latent semantic mining. Improving image classification accuracy is based on motivating the performance of the traditional back of visual words (BoW) framework. This is achieved by integrating color information into the classification system [28], image enhancement using histogram equalization method and image segmentation using improved Otsu algorithm [29]. A novel method is devised which depends on mixing images from two distinct classes while giving them random ratio, then feeding the training model with mixed image [30]. Furthermore, a combination of CNN and CapsNet has been proposed for classification of remote sensing image scene in [31]. The feature map is extracted using CNN. Then, to obtain the final classification result, the extracted feature map is used via CapsNet. Moreover, Yuan et al. [32] propose a rearranged feature for remote sensing image scene classification. First, extract features using CNN without fully connected layer, after that, features of images are clustered together to generate several collections.

More recently, high-precision scene classification methods have been summarized in [33]. To exploit more benefits of a pre-trained CNN for aerial scene classification, a multiscale pooling strategy (MSP) is proposed in [34] to achieve multiscale representation. MSP strategy extracts the local information from the last convolutional layer and exploited the fisher vector (FV) to generate the holistic scene representation. After that, a gated bidirectional network is proposed in [35] to integrate the hierarchical feature aggregation and the interference information elimination into an end-to-end network. First, the performance of each convolutional feature is quantitatively analyzed, and a superior combination of convolutional features are selected. Then, a bidirectional connection is proposed to hierarchically aggregate multilayer convolutional features. To reduce the number of parameters and achieve better classification performance in remote sensing image scene classification, He et al. [36] introduced a new skip-connected covariance (SCCov) network. The SCCov is to add skip connection and covariance pooling to CNNs. Multilayer feature fusion network (MF2Net) based on the VGGNet-16 model and fine-tuning techniques has been proposed in [37]. However, this method neglected some features like texture and directional information. To tackle this problem, a fast deep perception network (FDPResnet) based on residual neural network-101 (ResNet101) has been proposed in [38]. In addition, to automatically capture the latent ontological structure of VHR datasets, Wang et al. [39] proposed a multi-granularity canonical appearance pooling (MG-CAP) with the square root of the eigenvalue (Sqrt-E) instead of the original bilinear pooling. Yu et al. [40] utilized the fine-grained visual recognition to propose a bilinear CNN model, called BiMobileNet, for VHR scene classification. This model used the lightweight CNN (MobileNetv2) to extract deep and abstract image features and integrate them to achieve high VHR image classification performance. Finally, a multi-view feature learning network (MVFLN) is proposed in [41] to obtain three domain-specific features for the scene categorization task. Fully connected layers (FCLs) in the VGGNet are replaced by the channel-spatial branch and the other multiple metric branches.

The proposed model differs from the previous methods based on the generated features from pre-trained CNNs with a linear classifier. It expands the images with the augmentation techniques to ensure that the CNNs learn robust features. Moreover, to improve the accuracy of the classifier, the adaptive gradient algorithm (Adagrad) [42] is used with a multilayer perceptron (MLP) in the proposed model, which is named CNN-MLP. The major contributions of this work are three-fold:

- 1 An effective VHR image scene classification model named CNN-MLP is proposed in this paper. The data augmentation techniques are adopted during the training phase to avoid overfitting.
- 2 The proposed model uses a pre-trained CNNs model (i.e., a feature generation) for extracting the features from the original image and augmented ones in order to extract more detailed features for all VHR images.

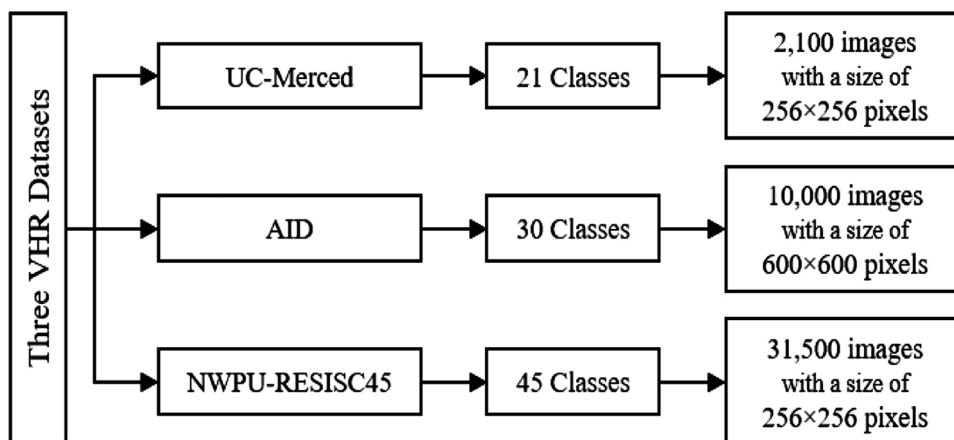


Fig. 1. Describe the three public VHR datasets selected.



Fig. 2. Sample images from UC-Merced dataset: (1) to (21) represent the twenty-one classes.

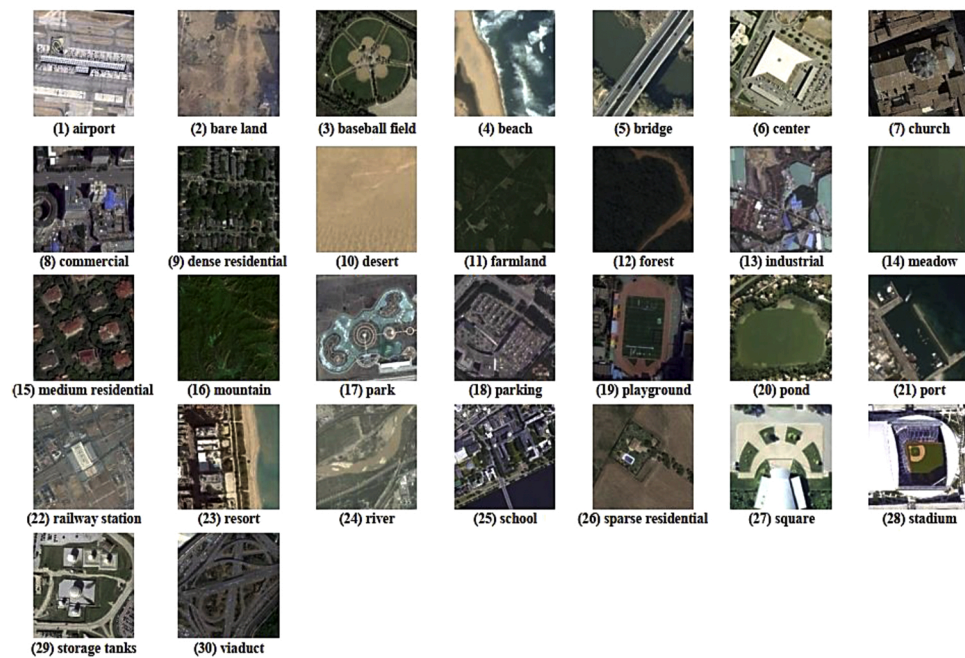


Fig. 3. Sample images from AID dataset: (1) to (30) represent the thirty classes.

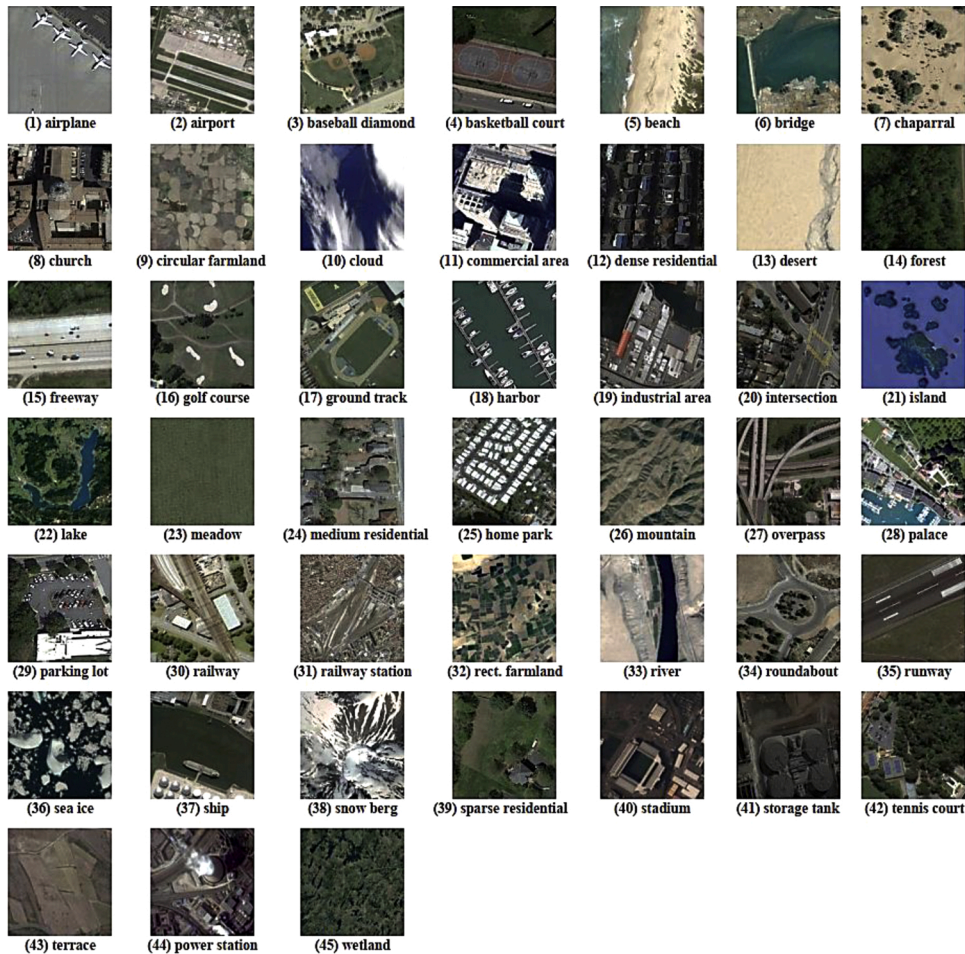


Fig. 4. Examples of NWPU-RESISC45 dataset: (1) to (45) represent the forty-five classes.

3 To classify the very high resolution (VHR) satellite image, this paper proposes an enhanced MLP classifier (i.e., a deep classifier) based on Adagrad optimizer [42]. The experimental results confirm that the proposed CNN-MLP model achieves high classification accuracies.

The rest of this paper is structured as follows. Section 2 describe the datasets, and then the framework of the proposed model is discussed. The experimental results are provided in Section 3, while Section 4 is dedicated to the conclusion of this paper.

2. Materials and proposed model

2.1. Materials

Three public datasets are used to verify the effectiveness of the proposed CNN-MLP model in VHR scene classification. 1) the UC-Merced dataset [43], 2) the aerial image dataset (AID) dataset [44], and 3) the NWPU-RESISC45 dataset [45]. Fig. 1 explains the details of the three datasets and explains what each dataset contains. Moreover, Fig. 2 shows some images representing different aerial scenes taken from UC-Merced dataset and Fig. 3 represents samples for the thirty classes of the AID dataset. Finally, Fig. 4 shows examples of the forty-five classes of the NWPU-RESISC45 dataset.

2.2. The proposed model

The overall workflow of the proposed CNN-MLP model is illustrated in Fig. 5. In the proposed classification model, there are three tasks need to be addressed: (1) Expanding the number of images by data augmentation, (2) Extracting the global features by using a

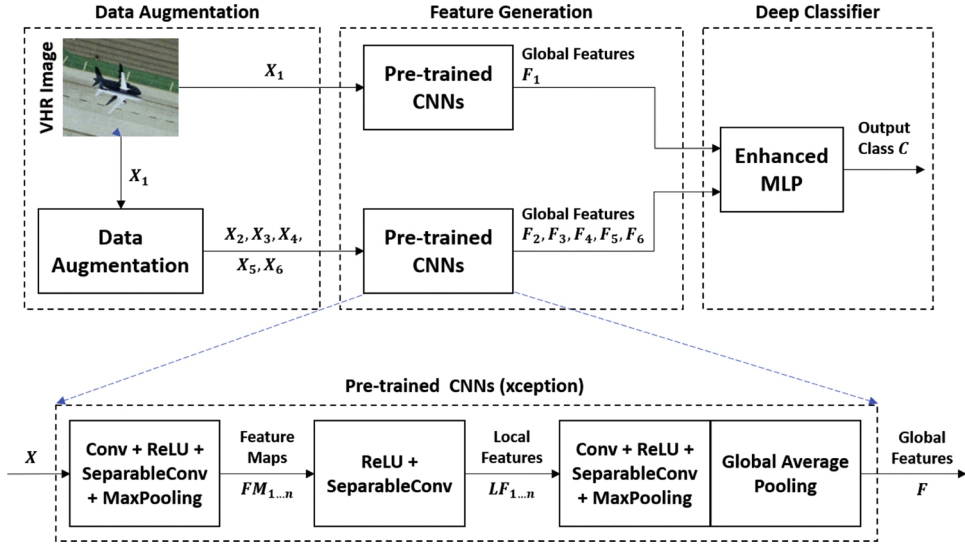


Fig. 5. A framework of the proposed CNN-MLP model for VHR imagery scene classification.

pre-trained CNNs model (Feature Generation), and (3) Enhanced MLP classifier (Deep Classifier). The pre-trained CNNs are used to extract the features from original VHR and augmented ones. Then, the extracted features are inserted into the enhanced MLP classifier. In the following subsections, each step of the proposed CNN-MLP model is discussed.

2.2.1. Data augmentation

Image data augmentation is a method that can be used to increase the number of images in the dataset by effecting transformations in the original images. Using Data Augmentation allows the proposed model to achieve better accuracy, especially for classes that are still mis-classified. As shown in Fig. 6, four images have been selected from the AID dataset. The distribution of pixels of these images is the same and all of them include several buildings and trees. However, there is irregularity of arrangement between building in school and in dense residential, where the buildings in ‘dense residential’ are properly organized and ordered.

In order for the model to be successful, large data is needed to train the model. The augmentation techniques can be used to generate various versions of images which are enhanced in model training while generalizing it. Different transformations on the training set are investigated in [46]. For each input image, there is generation of further 5 images that are rotated employing the image transformation equation which rotates each pixel (x, y) by multiplying it in the transformation matrix with pre-defined rotate angle θ in Eq. (1). These images are zoomed, horizontally flipped and vertically flipped, and this is performed. For instance, for a horizontal flip, the pixel situated at coordinate (x, y) will be situated at coordinate $(-x, y)$ in the new image using Eq. (2). In this equation, all x values are switched to their negative counterparts, while the y values remain the same. For a dataset of size N , the proposed method generates a dataset with size $5N$. Fig. 7 shows a sample of the produced images.

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad (1)$$

$$y = -f(x) \quad (2)$$

2.2.2. Convolution neural networks (CNNs)

Convolution neural networks (CNNs) are formed from a sequence of layers. The input map is moved through those layers till obtaining the output map. In the following subsections, a brief of those layers is given for illustration of the computation formula. Assume $X \in R^{h \times w \times c}$ (h : height, w : width, c : channel) is an RGB image. Every layer takes X and a set of parameters W as input and outputs a new image $Y \in R^{\tilde{h} \times \tilde{w} \times \tilde{c}}$, i.e., $Y = f(X, W)$.

First, a convolutional layer is the main layer of CNN. Learnable filters represent the parameter of this layer sliding the filter over all the input volume with pre-defined width and height. This produces an activation map representing the reaction of that filter at each spatial area. For computing the convolution of the input X with a bank of filters $W \in R^{\tilde{h} \times \tilde{w} \times \tilde{c} \times \tilde{c}}$ and adding a bias $\in R^{\tilde{c}}$, Eq. (3) is used.

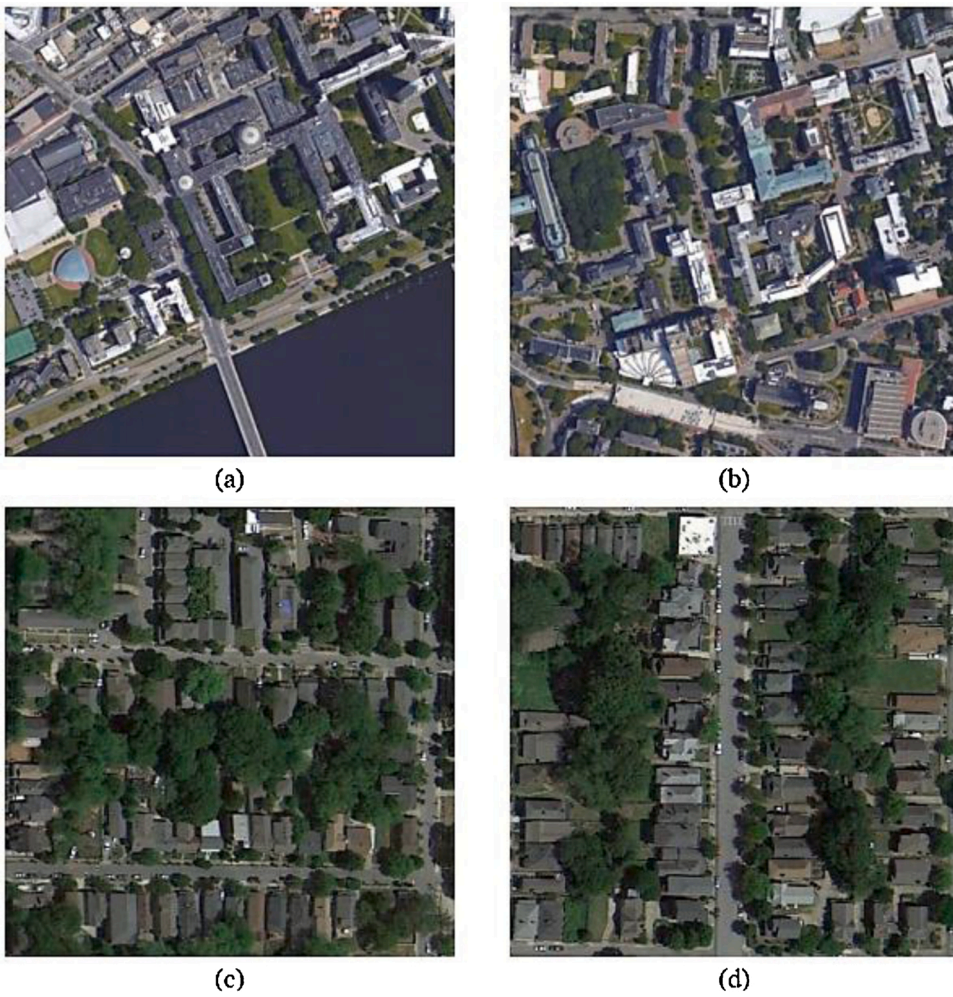


Fig. 6. Sample images from AID dataset. (a) and (b) are two images from school class. (c) and (d) two images from dense residential class.

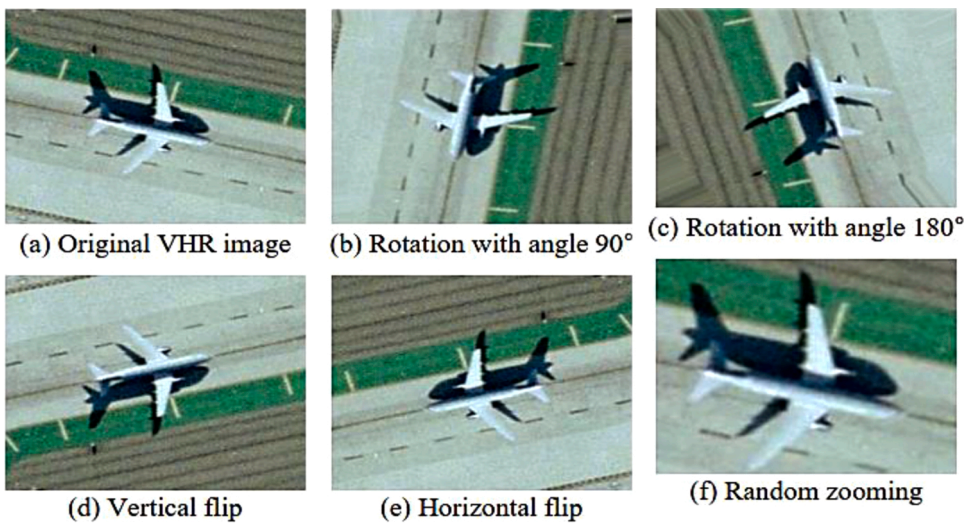


Fig. 7. Samples of data augmentation.

$$Y_{ijk} = f\left(b_k + \sum_{i=1}^{\tilde{h}} \sum_{j=1}^{\tilde{w}} \sum_{d=1}^c W_{ijk'} \times X_{i+i, j+j, d}\right) \quad (3)$$

Second, the max-pooling layer is used to decrease the parameters and computation within the network by reducing the size of impute shape. The output of the max-pooling layer is provided by maximum activation over shaped areas of size (h, w) . It computes the maximum response of each image channel in a $\tilde{h} \times \tilde{w}$ sub window, which acts as a subsampling operation. This can be written as Eq. (4) below:

$$Y_{ijk} = \max_{1 \leq i \leq \tilde{h}, 1 \leq j \leq \tilde{w}} X_{i+i, j+j, k} \quad (4)$$

Finally, fully connected layers (FCL) are a set of layers which merge the information extracted by previous layers (features). These layers take an input X , process it and the last FCL produce one dimensional vector of size equal the number of classes.

2.2.3. Feature extraction

In machine learning, features are the variables which an algorithm takes as input. Features can be specific parts of raw values from input or can be values obtained from that data. In the proposed model, we used a transfer learning technique to adapt a powerful pre-trained model to our VHR dataset. The xception [47] network is used as the feature extractor to automatically learn the feature descriptor for the VHR images. The xception pre-trained on the ImageNet dataset (A dataset with millions of images ranked in over 20, 000 classes). A smaller learning rate is used to fine-tune the convolution networks weights. After that, the logistic regression layer is freezed and only extract the global features F . Finally, the extracted features F are inserted into the MLP classifier.

The xception pre-trained CNN is applied to extract the features for the scene classification as shown in Fig. 5. For each VHR image, convolutional (conv) layers, Rectified Linear Unit (ReLU) as activation function, and max pooling layers generate the feature maps $FM_{1...n}$ for n patches from the VHR image. After that, separable convolutional layers and ReLU are used to compute the local maximum value of the region pixels (local features $LF_{1...n}$) in each channel of the generated feature maps. Finally, local features are then flattened into a feature vector and fed into the max pooling, conv, ReLU, and global average pooling layers to further extract the global features F of the whole image for the scene classification.

2.2.4. Classification using MLP

Investigation is conducted to exploit the advantages of MLP [48] with enhancements by adding a new dropout layer and using the Adagrad optimizer [42] for our proposed enhanced MLP. The classifier consists of three layers as shown in Fig. 8. It assumes that there are i features and k classes. The first layer normalizes all image features F_i . This normalization is a significant step to ensure that input data are distributed similarly. First, the mean of features of size M is obtained.

$$\mu = \frac{1}{M} \sum_{i=0}^M F_i \quad (5)$$

Then, variance is obtained.

$$\sigma^2 = \frac{1}{M} \sum_{i=0}^M (F_i - \mu)^2 \quad (6)$$

Finally, F_i to F_i^1 is normalized and a very small number ϵ is added to prevent the chance of a divide by zero error.

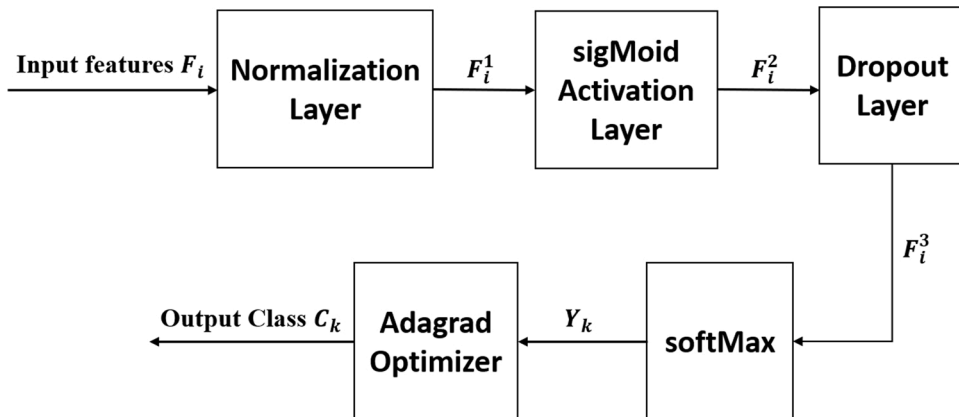


Fig. 8. The enhanced model of MLP classifier with i -input features and k -classes.

Table 1

Hyperparameters in the proposed model.

Hyperparameters	Value
Learning rate	0.01
Batch Normalization	True
Batch size	200
Loss function	cross-entropy
Activation function	ReLU + Softmax
Optimizer	Adagrad
Hidden layers	74
Number of filters	200
Kernel size	3×3
Dropout	True (0.4)
Feature scaling	True [0,1]

Table 2

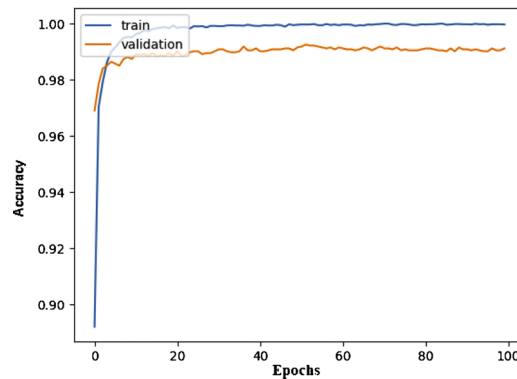
Benchmarking for the UC-Merced dataset with the literature.

Classification Method	Publication Year	Accuracy (%)
Fusion by addition [11]	2017	97.42
Resnet + WSPM-CRC [20]	2019	97.95
Two-Stream Fusion [15]	2018	98.02
GBNet [35]	2020	98.57
MSPs [34]	2019	98.60
ResNet_LGFFE [24]	2019	98.62
GCFs + LOFs [17]	2018	99.00
CTFCNN [22]	2019	99.02
Inception-v3-CapsNet [31]	2019	99.05
SCCov [36]	2019	99.05
MG-CAP (Sqrt-E) [39]	2020	99.10
BiMobileNet (MobileNetv2) [40]	2020	99.31
MVFLN + VGG-VD16 [41]	2020	99.52
CNN-MLP (ours)	2020	99.86

$$F_i^1 = \frac{F_i - \mu}{\sqrt{\sigma^2 + \varepsilon}} \quad (7)$$

This allows for a quicker convergence during network training. For mapping the result of each node between 0 and 1 in the second layer, the sigmoid activation function is used. The result of this layer is calculated using the following equation which takes F_i^1 and scales it between 0 and 1 in F_i^2 .

$$F_i^2 = \frac{1}{1 + e^{-F_i^1}} \quad (8)$$

**Fig. 9.** Plot for accuracy on the training and validation sets in UC-Merced dataset.

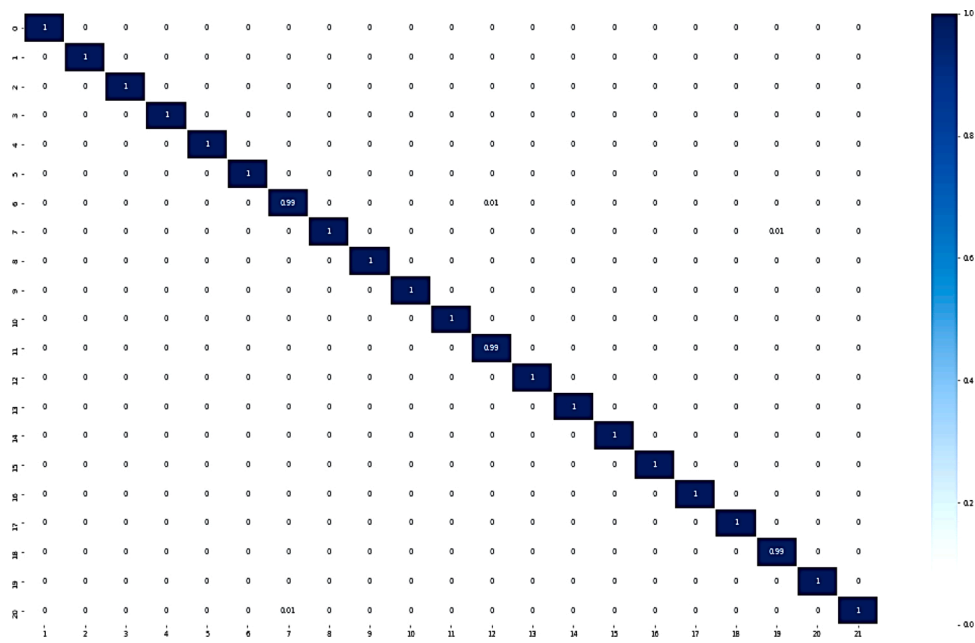


Fig. 10. Sample of a confusion matrix for UC-Merced dataset.

In the third layer, the dropout layer is used following the authors in [49]. Dropout is a regularization technique used to prevent shared modification on the trained data, which is reduced in model overfitting. This is followed by the use of softMax function that takes a vector of V as its input real numbers and normalizes it into a probability distribution consisting of V probabilities proportional to the exponential input number. Finally, the Adagrad algorithm is used for gradient-based optimization. Throughout the training, the main goal is to adapt the model's learning rate to how often a parameter is updated during training. m_t is defined as an accumulator of past gradient in Eq. (9) from which updates with different step sizes are computed for each parameter according to Eq. (10).

$$m_t = m_{t-1} + g_t^2 \quad (9)$$

$$\theta_{t+1} = \theta_t + \frac{\eta}{\sqrt{m_t} + \varepsilon} \ominus g_t \quad (10)$$

3. Experiments and analysis

Three experiments were implemented on the UC-Merced, AID, and NWPU-RESISC45 datasets to assess the performance of the classification models. The implementation details are presented in the first part of this section. Then, the description of our experiments and the results for each VHR dataset are discussed.

Table 3
Benchmarking for the AID dataset with the literature.

Classification Method	Publication Year	Accuracy (%)
salM ³ LBP-CLM [44]	2017	87.27
CaffeNet [44]	2017	87.33
ResNet_LGFEE [24]	2019	90.83
TEX-Net-LF [29]	2018	90.98
GBNet [35]	2020	92.20
CTFCNN [22]	2019	92.46
GCFs + LOFs [17]	2018	92.86
SCCov [36]	2019	93.12
MG-CAP (Sqrt-E) [39]	2020	93.52
Inception-v3-CapsNet [31]	2019	93.79
MF ² Net [37]	2020	93.82
MSPs [34]	2019	93.90
BiMobileNet (MobileNetv2) [40]	2020	95.07
MVFLN + VGG-VD16 [41]	2020	95.20
CNN-MLP (ours)	2020	98.10

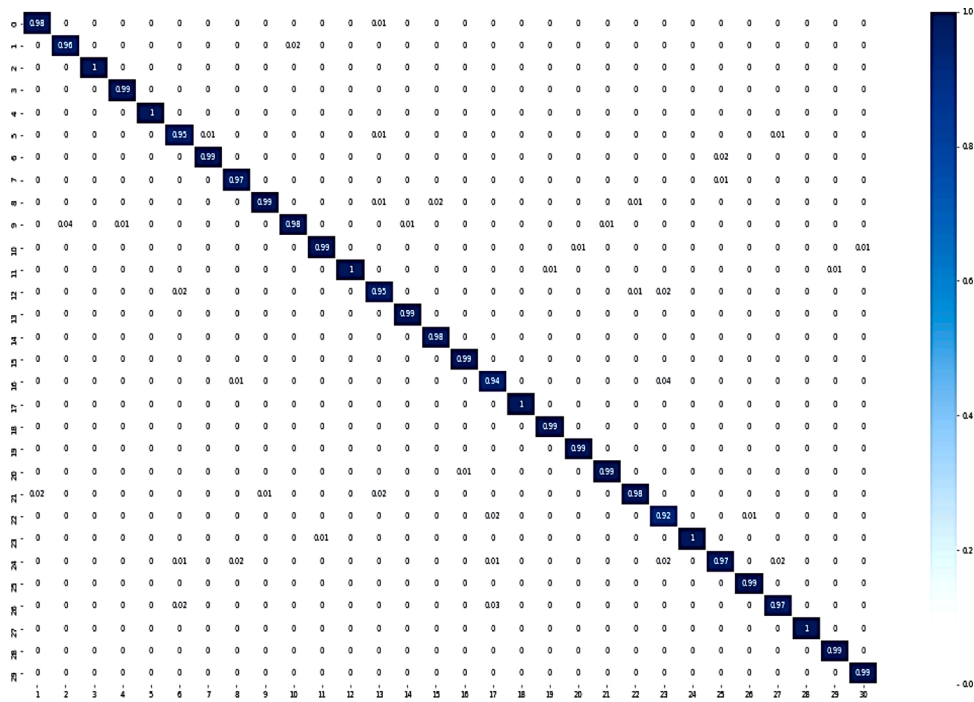


Fig. 11. Sample of a confusion matrix for AID dataset.

3.1. Implementation details

The proposed model expands the number of images by using data augmentation techniques. Then, the features of the images with pre-trained CNNs (i.e., xception [47]) are extracted. Finally, an enhanced MLP classifier is used for the classification step. All frameworks are implemented using Python and five tests are conducted. We conduct experiments using a PC with a 4-core Intel R Core TM i7–4790@3.60 GHz CPU and 16 GB memory. Table 1 shows the hyperparameters of the proposed CNN-MLP classification model.

In the experiments, the following steps have been applied to all used datasets. First, to benefit from transfer learning, in the proposed model, we used xception as the feature extractor to automatically learn the feature descriptor for the VHR images. All images are resized to 299×299 for the pre-trained CNNs. Second, the training ratio is set at 80 % for UC-Merced dataset, while the training ratio is set at 20 % for AID and NWPU-RESISC45 datasets. After that, we achieve the model accuracy by calculating the classification of each class C_i first as in Eq. (11), where Y_i and X_i represent the number of total and correct labels, respectively. Finally, the total accuracy is calculated by obtaining the mean of classes' accuracy as in Eq. (12).

$$C_i = X_i/Y_i \quad (11)$$

Table 4

Benchmarking for the NWPU-RESISC45 dataset with the literature.

Classification Method	Publication Year	Accuracy (%)
GoogLeNet [45]	2017	78.74
AlexNet [45]	2017	79.98
Two-Stream Fusion [18]	2018	83.34
BoCF (VGGNet-16) [12]	2017	84.49
Fine-Tuned GoogLeNet [45]	2017	86.20
Inception-v3-CapsNet [31]	2019	92.60
Fine-Tuned VGGNet-16 [45]	2017	90.54
SCCov [36]	2019	92.10
MF ² Net [37]	2020	92.73
MG-CAP (Sqrt-E) [39]	2020	93.06
BiMobileNet (MobileNetv2) [40]	2020	95.11
MVFLN + VGG-VD16 [41]	2020	94.46
FDPResNet [38]	2020	95.51
CNN-MLP (ours)	2020	97.40

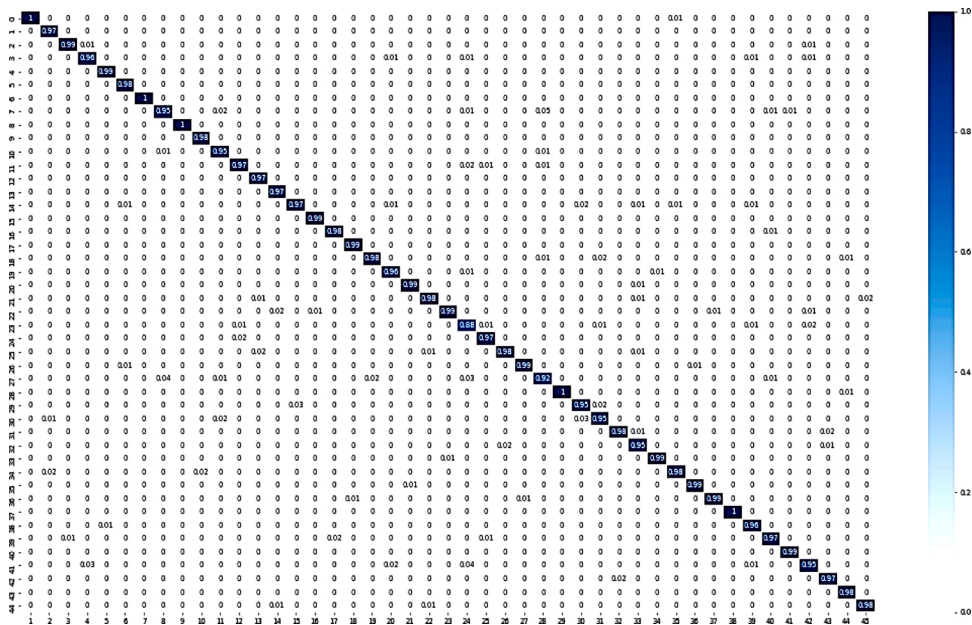


Fig. 12. Sample of a confusion matrix for NWPU-RESISC45 dataset.

$$\text{Total Accuracy} = \sum_{i=0}^N C_i / N \quad (12)$$

3.2. Experimental results and comparisons

The performance of the proposed CNN-MLP classification model in the selected VHR datasets is evaluated in the following subsections and is compared to the state-of-the-art scene classification methods in the same environment, conditions, and training/testing ratios. In total, the same experimental setup is run for 5 times. The mean of the five results is calculated to measure the performance of the proposed classification model. The accuracy comparison tables are presented, and the confusion matrices are drawn for all the selected VHR datasets. The experimental results confirm that the CNN-MLP model outperforms other state-of-the-art methods. Moreover, in the last subsection, the effectiveness of data augmentation used in our classification model is illustrated.

3.2.1. Classification of the UC-Merced dataset

To assess the classification performance, the results obtained from the experiments of the proposed CNN-MLP model were compared with those of other state-of-the-art methods in terms of VHR classification. In this subsection, the model was applied on the UC-Merced dataset. Table 2 compares the experimental results of CNN-MLP with those of the literatures. Evidently, the CNN-MLP model obtains remarkable classification results with an average accuracy of 0.34 % greater than the best model. Moreover, Fig. 9 gives the trend of accuracy while running the experiments, which shows the effectiveness of expanding the training images by using data augmentation techniques. Finally, the confusion matrix by CNN-MLP is shown in Fig. 10. From 0–20 represent the twenty-one classes in the UC-Merced dataset as shown in Fig. 2. From the confusion matrix, all the twenty-one categories have achieved accuracies more than 99 % in the classification.

3.2.2. Classification of the AID dataset

The second selected VHR dataset is the AID dataset. Sample images from AID dataset are shown in Fig. 3. The proposed model with CNN-MLP has achieved the best accuracy compared with the literature, as shown in Table 3. Experimental results on the AID dataset demonstrate that CNN-MLP increases the average accuracy by 4.28 %, 4.20 %, 3.03 % and 2.90 % over MF²Net [37], MSPs [34],

Table 5

Evaluation of the data augmentation used in the proposed CNN-MLP model on the three selected datasets.

Dataset	Accuracy (%)	
	With Data Augmentation	Without Data Augmentation
UC-Merced	99.86	97.20
AID	98.10	94.00
NWPU-RESISC45	97.40	93.70

BiMobileNet (MobileNetv2) [40], and MVFLN + VGG-VD16 [41] image classification models, respectively. Fig. 11 depict the confusion matrix generated from CNN-MLP for the AID dataset. The AID's classes in the confusion matrix are represented from 1 to 30 according to the class's name in Fig. 3. From Fig. 11, it is evident that twenty-two classes achieved accuracies greater than or equal 98 % in the classification.

3.2.3. Classification of the NWPU-RESISC45 dataset

Comparison between the proposed CNN-MLP model and the existing models using the NWPU-RESISC45 dataset is presented in Table 4. One can see that the CNN-MLP model has achieved leading classification results, with improvements of 1.89 % over the best model (i.e., FDRNet [38]). Furthermore, the confusion matrix result by CNN-MLP for the NWPU-RESISC45 dataset is shown in Fig. 12. In the confusion matrix, 0–44 represent the 45 classes in the dataset as shown in Fig. 4. One can see that, 34 among all the 45 classes have achieved classification accuracies greater than or equal 97 % in the confusion matrix.

3.2.4. The impact of data augmentation on the classification performance

To illustrate the gain brought by data augmentation within our CNN-MLP model, we have prepared a comparison in the accuracy in Table 5. In this comparison, the effectiveness of using data augmentation is shown. Results show improvement in accuracy by 2.66 %, 4.10 %, and 3.70 % on UC-Merced, AID, and NWPU-RESISC45, respectively.

4. Conclusion

An efficient deep neural network model, called CNN-MLP, is proposed for very high-resolution (VHR) imagery scene classification based on image features. In the proposed model, to improve the classification accuracy, an enhanced multilayer perceptron (MLP) depending on Adagrad optimizer is employed in the classification step. The proposed model expands the number of images in each class using data augmentation techniques and extracts the features from the image based on a pre-trained CNN model, then uses enhanced MLP in the classification of the extracted features. An evaluation of CNN-MLP model is conducted on the selected three VHR scene datasets and it is shown to achieve high classification accuracies of 99.86 %, 98.10 %, and 97.40 % for the UC-Merced, AID, and the NWPU-RESISC45 datasets, respectively. In future studies, investigating more sophisticated strategies is recommended for changing the MLP layers in order to improve accuracy.

Declaration of Competing Interest

None Declared.

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